Evaluation runs: chan (reliable-channel-aneris)

generated 2026-11-08 from brain-data e1e40f443 2026-11-08 13:42:54 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	8	882	conforms	0.1	2	1 h, 250 GB	1	1	brain03, 2026-11-02
3	48	23800	conforms	0.7	2	1 h, 250 GB	1	1	brain03, 2026-11-02
4	384	539540	conforms	7.0	4	1 h, 250 GB	1	1	brain03, 2026-11-02
5	3840	11217600	conforms	83.0	58	1 h, 250 GB	1	1	brain03, 2026-11-02
6	46080	221512004	conforms	2024.8	982	1 h, 250 GB	1	1	brain03, 2026-11-02
7	645120		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	8	704	362	violation	0.2	2	1 h, 250 GB	363	2897	brain03, 2026-11-02
3	48	20384	15064	violation	24.4	2	1 h, 250 GB	15065	723073	brain03, 2026-11-02
4	384	543060	462840	violation	1847.5	14	1 h, 250 GB	462841	177730561	brain03, 2026-11-02
5	3840			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

Not run (stateless). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag no-seqno, 5 no-seqno bag, 6 bag correct, 6 bag no-seqno, 6 bag no-seqno.t16.p.r3, 6 bag correct.t16.p.r3, 7 bag correct, 7 correct bag, 7 bag no-seqno, 7 bag no-seqno.t16.p.r3.

skipped: a smaller size had reached its limit: 6 no-seqno bag, 7 no-seqno bag, 8 correct bag, 8 no-seqno bag, 9 no-seqno bag, 9 correct bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	8	882	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-01
3	48	23800	conforms	1.0	2	1 h, 250 GB	brain02, 2026-11-01
4	384	539540	conforms	8.6	6	1 h, 250 GB	brain02, 2026-11-01
5	3840	11217600	conforms	97.0	60	48 h, 250 GB	brain05, 2026-11-02
6	46080	221512004	conforms	2462.0	984	48 h, 250 GB	brain05, 2026-11-02
7	645120	4226823920	conforms	53486.0	19194	48 h, 250 GB	brain01, 2026-11-03
11			🕓 (48 h)			48 h, 80 GB	pinky27, 2026-11-05

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
13			(48 h)			48 h, 80 GB	pinky31, 2026-11-05

stateful: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	8	704	362	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-01
3	48	20384	15064	violation	1.1	2	1 h, 250 GB	brain02, 2026-11-01
4	384	543060	462840	violation	9.0	8	1 h, 250 GB	brain02, 2026-11-01
5	3840	14061696	12864192	violation	129.7	139	48 h, 250 GB	brain05, 2026-11-02
6	46080	362542466	344704646	violation	4067.0	2208	48 h, 250 GB	brain05, 2026-11-02
7				(48 h)			48 h, 250 GB	brain04, 2026-11-02
8				(48 h)			48 h, 250 GB	brain02, 2026-11-03
13				(48 h)			48 h, 80 GB	pinky32, 2026-11-05

Not run (stateful). planned but never scheduled (grid points beyond the ladder): 6 no-seqno bag, 6 bag no-seqno.t16.p.r3, 6 bag correct.t16.p.r3, 7 correct bag, 7 bag no-seqno.t16.p.r3, 8 no-seqno bag, 8 correct bag, 9 no-seqno bag, 9 correct bag, 10 no-seqno bag, 10 correct bag, 11 no-seqno bag, 12 no-seqno bag, 12 correct bag.

left out by the table rules (ok): 5 correct bag, 5 no-seqno bag, 6 correct bag.

left out by the table rules (timeout): 6 no-seqno bag, 7 correct bag.

skipped: a smaller size had reached its limit: 7 no-seqno bag, 8 correct bag, 8 no-seqno bag, 9 no-seqno bag, 9 correct bag.

the tool ended with an error (exit-101): 8 bag correct.

left out by the table rules (invalid: foreign load): 9 bag no-seqno, 9 bag correct.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	882	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	48	23800	conforms	1.1	8	10 min, 16 GB	0.1	1.1	0.0	in-memory	brain05, 2026-11-02
4	384	539540	conforms	8.7	10	10 min, 16 GB	0.1	8.7	0.0	in-memory	brain05, 2026-11-02
5	3840	11217600	conforms	101.7		1 h, 250 GB	0.4	101.7	0.0	merge	brain05, 2026-11-03
6	46080	221512004	conforms	3464.8	982	1 h, 250 GB	1.0	3464.6	0.1	merge	brain05, 2026-11-03

offline: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	704	6	violation	0.1	9	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	48	20384	87	violation	0.9	8	10 min, 16 GB	0.1	0.9	0.0	in-memory	brain05, 2026-11-02
4	384	543060	1092	violation	10.8	26	10 min, 16 GB	0.1	10.8	0.0	in-memory	brain05, 2026-11-02
5	3840	14061696	14413	violation	122.5	78	1 h, 250 GB	0.4	122.1	0.4	merge	brain05, 2026-11-03

Not run (offline). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag no-seqno, 6 no-seqno bag, 6 bag correct, 6 bag no-seqno, 6 bag correct.p.r3, 6 bag no-seqno.p.r3, 7 bag correct, 7 bag no-seqno, 7 correct bag, 7 bag no-seqno.p.r3.

skipped: a smaller size had reached its limit: 6 no-seqno bag, 7 no-seqno bag, 7 correct bag, 8 no-seqno bag, 8 correct bag, 9 no-seqno bag, 9 correct bag.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	882	conforms	11.1	2035	10 min, 250 GB	0.1	11.1	0.1	merge	brain01, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	48	23800	conforms	84.4	4570	10 min, 250 GB	0.1	84.3	0.1	merge	brain01, 2026-11-03

cross offline, Java implementation against the Rust specification: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	704	6	violation	6.3	209	10 min, 250 GB	0.1	6.1	0.1	merge	brain01, 2026-11-03
3	48	20384	87	violation	42.7	4316	10 min, 250 GB	0.1	42.6	0.1	merge	brain01, 2026-11-03

Not run (cross offline, Java implementation against the Rust specification). planned but never scheduled (grid points beyond the ladder): 4 bag no-seqno, 4 bag correct, 5 bag correct, 5 bag no-seqno, 6 bag correct, 6 bag no-seqno, 7 bag correct, 7 bag no-seqno.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	882	conforms	4.1	175	10 min, 250 GB	4.0	0.1	0.0	merge	brain01, 2026-11-03
3	48	23800	conforms	4.4	172	10 min, 250 GB	4.3	1.1	0.1	merge	brain01, 2026-11-03
7	645120	4226823920	conforms	66165.1	36786	48 h, 250 GB	1049.7	66164.0	1.1	merge	brain08, 2026-11-08

cross offline, Rust implementation against the Java specification: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	704	6	violation	4.1	176	10 min, 250 GB	4.0	0.1	0.0	merge	brain01, 2026
3	48	20384	87	violation	4.4		10 min, 250 GB	4.3	0.9	0.1	merge	brain01, 2026
7	645120	9409563592	2723631	violation	145854.7	64717	48 h, 250 GB	1049.7	145742.0	112.7	merge	brain08, 2026

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 4 bag no-seqno, 4 bag correct, 5 bag correct, 5 bag no-seqno, 6 bag correct, 6 bag no-seqno.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	8	882	conforms	3.2	158	1 h, 250 GB		brain01, 2026-11-02
3	48	23800	conforms	4.7	170	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	8	704	362	violation	3.2	158	1 h, 250 GB	11	brain01, 2026-11-02
3	48	20384	15064	violation	4.6	170	1 h, 250 GB	1	brain01, 2026-11-02

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag no-seqno, 6 bag correct, 6 bag no-seqno, 7 bag correct, 7 bag no-seqno.

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	882	conforms	11.2	2035	10 min, 250 GB	11.1	0.1	0.1	merge	brain01, 2026-11-03
3	23800	23800	conforms	84.4	4570	10 min, 250 GB	84.3	1.1	0.1	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	704	704	conforms	6.2	209	10 min, 250 GB	6.1	0.1	0.1	merge	brain01, 2026-11-03
3	20384	20384	conforms	42.6	4316	10 min, 250 GB	42.6	0.9	0.1	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	704	6	violation	11.1	2035	10 min, 250 GB	11.1	0.1	0.0	merge	brain01, 2026-11-03
3	23800	20384	87	violation	84.4	4570	10 min, 250 GB	84.3	0.9	0.1	merge	brain01, 2026-11-03

Not run (twins offline, Rust implementation against its Java twin). planned but never scheduled (grid points beyond the ladder): 4 bag no-seqno, 4 bag correct, 5 bag correct, 5 bag no-seqno, 6 bag correct, 6 bag no-seqno.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	882	conforms	11.3	2035	10 min, 250 GB	0.1	11.1	0.2	merge	brain01, 2026-11-03
3	23800	23800	conforms	84.4	4570	10 min, 250 GB	1.1	84.3	0.1	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	704	704	conforms	6.2	209	10 min, 250 GB	0.1	6.1	0.0	merge	brain01, 2026-11-03
3	20384	20384	conforms	42.8		10 min, 250 GB	0.9	42.6	0.2	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	704	6	violation	6.2	209	10 min, 250 GB	0.1	6.1	0.1	merge	brain01, 2026-11-03
3	23800	20384	87	violation	42.8	4316	10 min, 250 GB	1.1	42.6	0.2	merge	brain01, 2026-11-03

Not run (twins offline, Java implementation against its Rust twin). planned but never scheduled (grid points beyond the ladder): 4 bag no-seqno, 4 bag correct, 5 bag correct, 5 bag no-seqno, 6 bag correct, 6 bag no-seqno.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	882	conforms	8.5	260	10 min, 250 GB	8.5	0.1	0.0	merge	brain01, 2026-11-03
3	23800	23800	conforms	52.3	5014	10 min, 250 GB	52.0	1.1	0.3	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	704	704	conforms	6.0	238	10 min, 250 GB	5.9	0.1	0.0	merge	brain01, 2026-11-03
3	20384	20384	conforms	41.0	4368	10 min, 250 GB	40.9	0.9	0.1	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	704	6	violation	8.6	260	10 min, 250 GB	8.5	0.1	0.1	merge	brain01, 2026-11-03
3	23800	20384	87	violation	52.1	5014	10 min, 250 GB	52.0	0.9	0.1	merge	brain01, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). planned but never scheduled (grid points beyond the ladder): 4 bag no-seqno, 4 bag correct, 5 bag correct, 5 bag no-seqno, 6 bag correct, 6 bag no-seqno.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	882	conforms	8.6	260	10 min, 250 GB	0.1	8.5	0.1	merge	brain01, 2026-11-03
3	23800	23800	conforms	52.1	5014	10 min, 250 GB	1.1	52.0	0.1	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	704	704	conforms	6.0	238	10 min, 250 GB	0.1	5.9	0.1	merge	brain01, 2026-11-03
3	20384	20384	conforms	40.9	4368	10 min, 250 GB	0.9	40.9	0.0	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	704	6	violation	5.9	238	10 min, 250 GB	0.1	5.9	0.0	merge	brain01, 2026-11-03
3	23800	20384	87	violation	41.1	4368	10 min, 250 GB	1.1	40.9	0.2	merge	brain01, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). planned but never scheduled (grid points beyond the ladder): 4 bag no-seqno, 4 bag correct, 6 bag no-seqno.

skipped: a smaller size had reached its limit: 5 bag no-seqno, 5 bag correct, 6 bag correct.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	882	882	conforms	9.9	280	1 h, 250 GB		brain01, 2026-11-03
3	23800	23800	conforms	64.4	5055	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	704	704	conforms	9.2	236	1 h, 250 GB		brain01, 2026-11-03
3	20384	20384	conforms	57.7	4930	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	882	704	362	violation	10.0	236	1 h, 250 GB	95	brain01, 2026-11-03
3	23800	20384	15064	violation	57.6	4658	1 h, 250 GB	142	brain01, 2026-11-03

Not run (context-free twins online, Java port (JMC) against the Rust implementation's export). planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno, 5 bag correct, 6 bag no-seqno, 6 bag correct.

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	882	882	conforms	8.0	258	1 h, 250 GB		brain01, 2026-11-02
3	23800	23800	conforms	68.4	6167	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	704	704	conforms	5.1	295	1 h, 250 GB		brain01, 2026-11-02
3	20384	20384	conforms	55.9	6288	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	882	704	362	violation	8.0	258	1 h, 250 GB	10	brain01, 2026-11-02
3	23800	20384	15064	violation	68.1	6167	1 h, 250 GB	7	brain01, 2026-11-02

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).
skipped: a smaller size had reached its limit: 5 bag no-seqno, 5 bag correct, 6 bag no-seqno, 6 bag correct.
planned but never scheduled (grid points beyond the ladder): 6 bag no-seqno.